

Air University



Operating Systems (Lab)

Year 2026

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Department of Cyber Security

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Assignment # 01

Question 1: Structured Data Program (VI + Compilation)

Create file:

Navigate into the assignment_01 directory and use touch and vi to create a new c++ source file named record.cpp.

```
usama@ubuntu:~/Desktop$ cd Assignment_01
usama@ubuntu:~/Desktop/Assignment_01$ touch record.cpp
usama@ubuntu:~/Desktop/Assignment_01$ vi record.cpp
```

Code editor:

This program code prompts for student details and marks, calculates the total and average, and uses an if-else block to determine a "Pass" or "Fail" status.

```
#include <iostream>
#include <cstring>

using namespace std;

int main()
{
    string name;
    cout << "Enter Name: ";
    getline(cin, name);

    int roll_no;
    cout << "Enter Roll No: ";
    cin >> roll_no;

    int subject_1, subject_2, subject_3;
    cout << "Enter Subject 1 Marks: ";
    cin >> subject_1;
    cout << "Enter Subject 2 Marks: ";
    cin >> subject_2;
    cout << "Enter Subject 3 Marks: ";
    cin >> subject_3;

    int total_Marks;
    total_Marks = subject_1 + subject_2 + subject_3;

    int average;
    average = total_Marks / 3;

    cout << "\n----- Student Report -----";
    cout << "\nName: " << name << "\nRoll No: " << roll_no;
    cout << "\nTotal: " << total_Marks << "\nAverage: " << average << "%";

    if (average < 50)
    {
        cout << "\nStatus: Fail";
    }
    else
    {
        cout << " \nStatus: Pass";
    }
    return 0;
}
```

Compilation & Execution:

After compiling with g++, used the ./record executable, which correctly processes user input to generate a formatted "Student Report" with a passing status.

```
usama@ubuntu:~/Desktop/Assignment_01$ g++ record.cpp -o record
usama@ubuntu:~/Desktop/Assignment_01$ ./record
Enter Name: Usama Arshad
Enter Roll No: 247141
Enter Subject 1 Marks: 75
Enter Subject 2 Marks: 85
Enter Subject 3 Marks: 70

----- Student Report -----
Name: Usama Arshad
Roll No: 247141
Total: 230
Average: 76%
Status: Pass
```

Question 2: Argument-Based Calculator

Create file:

Initialized the calculator task by creating the calc.cpp file and opening the vi editor in assignment directory.

```
usama@ubuntu:~/Desktop/Assignment_01$ touch calc.cpp
usama@ubuntu:~/Desktop/Assignment_01$ vi calc.cpp
```

Code Editor:

This is the complete source code for calc.cpp, featuring atoi for argument conversion and a case-like if-else structure for math operations.

```
#include <iostream>
#include <string>
#include <cstdlib>
using namespace std;

int main(int argc, char *argv[])
{
    // Part 1: Validate argument count (Program name + 3 inputs = 4)
    if (argc != 4)
    {
        cout << "Usage: ./calc num1 operator num2" << endl;
        cout << "Example: ./calc 10 + 5" << endl;
        return 1; // Exit with error code
    }

    // Part 2: Convert arguments to usable types
    int num1 = atoi(argv[1]);
    string op = argv[2];
    int num2 = atoi(argv[3]);

    // Part 3: Perform calculation based on operator
    if (op == "+")
    {
        cout << "Result: " << (num1 + num2) << endl;
    }
    else if (op == "-")
    {
        cout << "Result: " << (num1 - num2) << endl;
    }
    else if (op == "x" || op == "**")
    {
        // Note: '*' is a wildcard in Linux, so 'x' is safer for testing
        cout << "Result: " << (num1 * num2) << endl;
    }
    else if (op == "/")
    {
        // Part 4: Handle division by zero
        if (num2 == 0)
        {
            cout << "Error: Division by zero!" << endl;
        }
        else
        {
            cout << "Result: " << (num1 / num2) << endl;
        }
    }
    else
    {
        cout << "Error: Invalid operator. Use +, -, x, /" << endl;
    }

    return 0;
}
```

Compilation & Execution:

After compiling the calculator, I tested it with multiple operations (division, addition, subtraction, multiplication) and verified the handling of the "Division by zero" error.

```
usama@ubuntu:~/Desktop/Assignment_01$ g++ calc.cpp -o calc
usama@ubuntu:~/Desktop/Assignment_01$ ./calc 6 / 2
Result: 3
usama@ubuntu:~/Desktop/Assignment_01$ ./calc 3 + 2
Result: 5
usama@ubuntu:~/Desktop/Assignment_01$ ./calc 8 - 4
Result: 4
usama@ubuntu:~/Desktop/Assignment_01$ ./calc 3 x 4
Result: 12
usama@ubuntu:~/Desktop/Assignment_01$ ./calc 3 / 0
Error: Division by zero!
```

Question 3: Argument Analyzer

Create file:

Used the touch command to create an empty analyze.cpp file and the vi editor to open it for coding.

```
usama@ubuntu:~/Desktop/Assignment_01$ touch analyze.cpp
usama@ubuntu:~/Desktop/Assignment_01$ vi analyze.cpp
```

Code Editor:

This is the C++ source code that uses a for loop to iterate through argv, tracking the longest string and summing the total character count.

```
#include <iostream>
#include <string>
#include <vector>
using namespace std;

int main(int argc, char *argv[])
{
    // Part 1: Print program name (argv[0]) and total arguments
    cout << "Program Name: " << argv[0] << endl;
    cout << "Total Number of Arguments: " << argc - 1 << endl;

    if (argc < 2)
    {
        cout << "No extra arguments provided." << endl;
        return 0;
    }

    string longestArg = argv[1];
    int totalCharacters = 0;

    // Part 2: Loop through arguments (starting from index 1)
    for (int i = 1; i < argc; ++i)
    {
        string currentArg = argv[i];

        // Accumulate total characters
        totalCharacters += currentArg.length();

        // Check if current argument is longer than the one we've stored
        if (currentArg.length() > longestArg.length())
        {
            longestArg = currentArg;
        }
    }

    // Part 3: Display results
    cout << "Longest Argument: " << longestArg << endl;
    cout << "Total Characters across all arguments: " << totalCharacters << endl;

    return 0;
}
```

Compilation & Execution:

Successfully compiled the code with g++ and executed it with various arguments to verify the logic for different input lengths.

```
usama@ubuntu:~/Desktop/Assignment_01$ g++ analyze.cpp -o analyze
usama@ubuntu:~/Desktop/Assignment_01$ ./analyze
Program Name: ./analyze
Total Number of Arguments: 0
No extra arguments provided.
usama@ubuntu:~/Desktop/Assignment_01$ ./analyze OS Lab Ubuntu
Program Name: ./analyze
Total Number of Arguments: 3
Longest Argument: Ubuntu
Total Characters across all arguments: 11
usama@ubuntu:~/Desktop/Assignment_01$ ./analyze OS Lab Assignment_01
Program Name: ./analyze
Total Number of Arguments: 3
Longest Argument: Assignment_01
Total Characters across all arguments: 18
```

THE END
